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Senior Data Scientist

Government Digital Service

Apply before 11:55 pm on Sunday 22nd November 2026

Government Digital Service

Reference number

489652

Salary

£58,275 - £73,875

£58,275 - £73,875 for National Based on capability. The base salary of this grade is £58,275. Offers made above this will be made up with a specialist pay allowance.

Job grade

Grade 7

Contract type

Permanent

Business area

GDS - Products & Services

Type of role

Digital
Information Technology

Working pattern

Flexible working, Full-time, Job share, Part-time, Compressed hours

Number of jobs available

1

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Location

Bristol, Manchester

About the job

Job summary

The Government Digital Service (GDS) is the digital centre of government - we are responsible for setting, leading and delivering the vision for a modern digital government.

Our priorities are to drive a modern digital government, by:

1. joining up public sector services
2. harnessing the power of AI for the public good
3. strengthening and extending our digital and data public infrastructure
4. elevating leadership and investing in talent
5. funding for outcomes and procuring for growth and innovation
6. committing to transparency and driving accountability

We are home to the Incubator for Artificial Intelligence (I.AI), the world-leading GOV.UK and at the forefront of coordinating the UK s geospatial strategy and activity. We lead the Government Digital and Data function and champion the work of digital teams across government.

We re part of the Department for Science, Innovation and Technology (DSIT) and employ more than 1,000 people all over the UK, with hubs in Manchester, London and Bristol.

The Government Digital Service is where talent translates into impact. From your first day, you ll be working with some of the world s most highly-skilled digital professionals, all contributing their knowledge to make change on a national scale.

Join us for rewarding work that makes a difference across the UK. You'll solve some of the nation s highest-priority digital challenges, helping millions of people access services they need

Former DSIT recruitment campaigns are continuing as usual, but candidates should be aware that following the Government s announcement on the changes to some civil service departments, roles

will be subject to the machinery of government moves and will ultimately be in one of the new departments. We will provide more information if you are selected for a role. This work remains of high importance to the civil service, and we thank you for your continued interest.

Job description

GDS is helping government get real, responsible value from AI. As a Senior Data Scientist in the GDS Product and Services AI programme, you'll turn fast-moving AI capability into tested, trustworthy services that improve how people experience government.

You'll work with product, delivery, user research, design and engineering colleagues to find where AI can help, check whether the underlying data and content are good enough, and design and run pilots that prove through rigorous testing what works, what doesn't and why. That includes building evaluation methods for accuracy, safety and user value, and testing agentic approaches in controlled pilots before they reach the public. Your findings will shape how GDS and the rest of government adopt AI.

About the team

The AI programme sits within the GDS Products and Services directorate. It supports government to adopt AI safely, effectively and at scale. The team builds and improves GOV.UK Chat, which has been through large-scale public pilots, red-team and security testing, and a launch in the GOV.UK app. It is also experimenting with agentic AI, including how an assistant could take simple actions on a person's behalf. The team trials new tools, builds shared capability and guidance, and works with departments on practical use cases that meet public sector standards for security, transparency and ethics.

You can read about some of our work here:

- https://gds.blog.gov.uk/2026/05/14/gov-uk-chat-launches/
- https://insidegovuk.blog.gov.uk/2026/05/15/developing-gov-uk-chat-our-data-science-and-ai-engineering-journey/

As a Senior Data Scientist, you ll:

- lead technical work on evaluation and experimentation within the AI programme, working with product, delivery, user research, design and engineering colleagues to build AI products that meet user needs
- prototype and assess new and emerging AI tools, including agent-based systems, and make evidence-based recommendations on whether they're worth adopting for government use cases
- shape problems with product and technical leads, helping to define scope and success measures, then design and run experiments and pilots using generative AI, agentic approaches and automation
- design robust, repeatable evaluation frameworks that show how well a tool or solution performs, where it fails and what risks it carries, including red-teaming and guardrails
- judge whether datasets and content are suitable and representative to use for a given purpose, and advise on gaps and remedies
- write and review secure, maintainable, well-tested code, and promote good engineering practice in data science work
- explain results, trade-offs and recommendations to audiences from engineers to senior officials, so decisions rest on evidence
- apply government standards on data protection, security, transparency and AI ethics, and contribute to responsible AI practice in the team and the wider Data and AI community
- mentor and coach data scientists and share good practice, helping to raise capability in the team

Person specification

We re interested in people who:

- a track record of applying data science and generative AI methods to real operational or user problems, working in multidisciplinary teams
- sound analytical rigour: you've evaluated models, tools or experiments, including LLM or RAG systems, using methods such as benchmark

datasets, human or LLM-based grading and red-teaming, and reached clear, defensible conclusions that shape product delivery

the ability to explain technical findings, limitations and uncertainty to non-technical decision-makers

good judgement on responsible use of data and AI, including governance, privacy, security and ethics

strong engineering habits: Python, clean and tested code, CI/CD, version control, and practical use of AI-assisted coding tools to improve quality and speed

hands-on experience building LLM-powered applications and AI agents, including retrieval-augmented generation (RAG), with an understanding of embeddings, vector search and retrieval quality, using modern AI development frameworks (e.g., Amazon Strands, LangChain, LangGraph), and cloud AI platforms (preferably AWS Bedrock)

understanding of agentic design patterns, such as tool and function calling, memory and multi-agent set-ups, and of how to test them safely.

experience in the public sector, regulated environments, research or fast-moving start-ups.

context engineering experience across commercial and open-source models.

Benefits

There are many benefits of working at GDS, including:

flexible hybrid working with flexi-time and the option to work part-time or condensed hours

a Civil Service Pension with an average employer contribution of 28.97%

25 days of annual leave, increasing by a day each year up to a maximum of 30 days

an extra day off for the King s birthday

an in-year bonus scheme to recognise high performance

career progression and coaching, including a training budget for personal development

- a focus on wellbeing with access to an employee assistance programme
- job satisfaction from making government services easier to use and more inclusive for people across the UK
- advances on pay, including for travel season tickets
- death in service benefits
- cycle to work scheme and facilities
- access to children's holiday play schemes across different locations in central London
- access to an employee discounts scheme
- 10 learning days per year
- volunteering opportunities (5 special leave days per year)
- access to a suite of learning activities through Civil Service learning

Any move to Government Digital Service from another employer will mean you can no longer access childcare vouchers. This includes moves between government departments. You may however be eligible for other government schemes, including Tax Free Childcare. Determine your eligibility at https://www.childcarechoices.gov.uk

Office attendance

The Department operates a discretionary hybrid working policy, which provides for a combination of working hours from your place of work and from your home in the UK. The current expectation for staff is to attend the office or non-home based location for 40-60% of the time over the accounting period.

DSIT does not normally offer full home working (i.e. working at home); but we do offer a variety of flexible working options (including occasionally working from home).

GDS offers hybrid working for all employees. This means that everyone does some working from home and also spends some time in their local office. You ll agree to your hybrid working arrangement with your line manager in line with your preferences and business needs.

Any move to Government Digital Service from another employer will mean you can no longer access childcare vouchers. This includes moves between

government departments. You may however be eligible for other government schemes, including Tax Free Childcare. Determine your eligibility at https://www.childcarechoices.gov.uk

Things you need to know

Artificial intelligence

Artificial intelligence can be a useful tool to support your application, however, all examples and statements provided must be truthful, factually accurate and taken directly from your own experience. Where plagiarism has been identified (presenting the ideas and experiences of others, or generated by artificial intelligence, as your own) applications may be withdrawn and internal candidates may be subject to disciplinary action. Please see our candidate guidance (opens in a new window) for more information on appropriate and inappropriate use.

Selection process details

The standard selection process for roles at GDS consists of:

- a simple application screening process - We only ask for a CV and cover letter of up to 750 words. Important tip - please ensure that your cover letter includes how you meet the skills and experience listed in the person specification section above
- A 60-minute video interview: behaviour-based questions plus a technical discussion.

Depending on how many applications we get, there might also be an extra stage before the video interview, for example a phone interview or a technical exercise.

While we value the use of AI technology to enhance our daily work, we also value the personal touch and urge applicants to write cover letters without the use of AI to emphasise their own unique experiences.

In the event we receive a high volume of applications, we will conduct the initial sift against the lead criteria which is:

- a track record of applying data science and generative AI methods to real operational or user problems, working in multidisciplinary team

In the Civil Service, we use Success Profiles to evaluate your skills and ability. This gives us the best possible chance of finding the right person for the job, increases performance and improves diversity and inclusivity. We ll be assessing your technical abilities, skills, experience and behaviours that are relevant to this role.

For this role we ll be assessing you against the following Civil Service Behaviours:

- working together
- changing and improving
- making effective decisions
- developing self and others
- seeing the bigger picture
- communicating and influencing

We ll also be assessing your experience and specialist technical skills against the following skills defined in the Government Digital and Data Capability Framework for the Data Scientist role:

- understanding product delivery
- programming and build (data science)
- data science innovation
- delivering business impact
- developing data science capabilities
- data engineering
- applied maths, statistics and scientific practices
- data ethics and privacy

Want to know more about who Government Digital and Data are? Click Here

Recruitment Timeline

Sift completion: 30/11/2026

Panel interviews: W/C 7/12/2026

Candidates that do not pass the interview but have demonstrated an acceptable standard may be considered for similar roles at a lower grade.

A reserve list will be held for a period of 12 months, from which further appointments can be made.

The Civil Service is committed to attract, retain and invest in talent wherever it is found. To learn more please see the Civil Service People Plan and the Civil Service D&I Strategy.

Sponsorship

DSIT cannot offer Visa sponsorship to candidates through this campaign. DSIT holds a Visa sponsorship licence but this can only be used for certain roles and this campaign does not qualify.

Location

Whilst this role's location is Bristol / Manchester, if you are a current Civil servant working in DSIT and based in London, you are also able to apply for this role. If you are successful, you will retain your London location and pay.

Security

Successful candidates must pass a disclosure and barring security check.

Please note that this role requires SC clearance, which would normally need 5 years UK residency in the past 5 years. This is not an absolute requirement, but supplementary checks may be needed where individuals have not lived in the UK for that period. This may mean your security clearance (and therefore your appointment) will take longer or, in some cases, not be possible.

For meaningful checks to be carried out, you will need to have lived in the UK for a sufficient period of time, to enable appropriate checks to be carried out and produce a result which provides the required level of assurance. Whilst a lack of UK residency in itself is not necessarily a bar to a security clearance, and expectation of UK residency may range from 3 to 5 years. Failure to meet the residency requirements needed for the role may result in the withdrawal of provisional jobs offers.

Feedback will only be provided if you attend an interview or assessment.

Security

Successful candidates must undergo a basic (or equivalent) criminal record check.

Successful candidates must meet the security requirements before they can be appointed. The level of security needed is security check (opens in a new window).

See our vetting charter (opens in a new window).

People working with government assets must complete baseline personnel security standard (opens in new window) checks.

Nationality requirements

This job is broadly open to the following groups:

- UK nationals
- nationals of the Republic of Ireland
- nationals of Commonwealth countries who have the right to work in the UK
- nationals of the EU, Switzerland, Norway, Iceland or Liechtenstein and family members of those nationalities with settled or pre-settled status under the European Union Settlement Scheme (EUSS) (opens in a new window)
- nationals of the EU, Switzerland, Norway, Iceland or Liechtenstein and family members of those nationalities who have made a valid application

- for settled or pre-settled status under the European Union Settlement Scheme (EUSS)
- individuals with limited leave to remain or indefinite leave to remain who were eligible to apply for EUSS on or before 31 December 2020
- Turkish nationals, and certain family members of Turkish nationals, who have accrued the right to work in the Civil Service

Further information on nationality requirements (opens in a new window)

Working for the Civil Service

The Civil Service Code (opens in a new window) sets out the standards of behaviour expected of civil servants.

We recruit by merit on the basis of fair and open competition, as outlined in the Civil Service Commission's recruitment principles (opens in a new window).

The Civil Service embraces diversity and promotes equal opportunities. As such, we run a Disability Confident Scheme (DCS) for candidates with disabilities who meet the minimum selection criteria.

The Civil Service also offers a Redeployment Interview Scheme to civil servants who are at risk of redundancy, and who meet the minimum requirements for the advertised vacancy.

Diversity and Inclusion

The Civil Service is committed to attract, retain and invest in talent wherever it is found. To learn more please see the Civil Service People Plan (opens in a new window) and the Civil Service Diversity and Inclusion Strategy (opens in a new window).

Apply and further information

This vacancy is part of the Great Place to Work for Veterans (opens in a new window) initiative.

The Civil Service welcomes applications from people who have recently left prison or have an unspent conviction. Read more about prison leaver recruitment (opens in new window).

Once this job has closed, the job advert will no longer be available. You may want to save a copy for your records.

Contact point for applicants

Job contact :

Name :	gdsrecruitment@dsit.gov.uk
Email :	gdsrecruitment@dsit.gov.uk

Recruitment team

Email :	gdsrecruitment@dsit.gov.uk

Further information

If you feel your application has not been treated in accordance with the Recruitment Principles and you wish to make a complaint, you should contact gds-complaints@dsit.gov.uk in the first instance.

If you are not satisfied with the response you receive you can contact the Civil Service Commission by email: info@csc.gov.uk Or in writing: Civil Service Commission, Room G/8 1 Horse Guards Road, London, SW1A 2HQ.

Civil Service Commission

disability confident EMPLOYER